

# Technical Training - C# & .NET

Introductory support material created from the supplied 13-week training plan. It gives each week a clear session flow, outcomes and lightweight practice.

## Week at a glance

| Session   | Focus                       | Expected outcome                                                                                 |
|-----------|-----------------------------|--------------------------------------------------------------------------------------------------|
| Session 1 | Object-Oriented Programming | Cover classes, objects, encapsulation, inheritance, polymorphism and interfaces.                 |
| Session 2 | .NET Fundamentals           | Understand the .NET runtime, project structure, packages and application execution.              |
| Session 3 | Application Structure       | Organize code into clear layers/components and understand separation of responsibilities.        |
| Session 4 | Development Practices       | Review debugging, configuration, dependency use, clean code and maintainable development habits. |

## Session notes

### Session 1: Object-Oriented Programming

Cover classes, objects, encapsulation, inheritance, polymorphism and interfaces.

**Quick check:** Explain the idea in your own words and identify one practical place where it could be used.

### Session 2: .NET Fundamentals

Understand the .NET runtime, project structure, packages and application execution.

**Quick check:** Explain the idea in your own words and identify one practical place where it could be used.

### Session 3: Application Structure

Organize code into clear layers/components and understand separation of responsibilities.

**Quick check:** Explain the idea in your own words and identify one practical place where it could be used.

### Session 4: Development Practices

Review debugging, configuration, dependency use, clean code and maintainable development habits.

**Quick check:** Explain the idea in your own words and identify one practical place where it could be used.

## Practice and reflection

These are lightweight practice prompts for the onboarding prototype, not an official assessment.

1. Write five key terms from this week and explain each in one sentence.
2. Draw a simple flow showing how the main concepts connect.
3. List two questions you would ask the trainer before using these concepts in a real project.
4. Identify one practical example from a utility, software system, workplace process or assigned role.
5. Write an end-of-week summary: what you understood, what remains unclear, and what you want to practise next.

## Completion checklist

- ☐ I attended/reviewed the planned sessions.
- ☐ I can explain the week's main concepts at a basic level.
- ☐ I completed at least one practice/reflection activity.
- ☐ I recorded questions or gaps for follow-up.
- ☐ I am ready to proceed to the next onboarding chapter.

**Note:** This is prototype support material. The supplied training plan remains the source for the official week/topic sequence; trainers may provide deeper or updated material.